

Project Proposal: Audio-JEPA for Expressive, Low-Resource Arabic Text-to-Speech

KASP 2026 AI Project | Presented to Mentor: Kerven Durdymyradov

Team: Abdelrahman, Hassan, Omar

Research-backed Arabic TTS prototype

Evaluate whether self-supervised Audio-JEPA-style representations can improve Arabic TTS under limited paired-data conditions.

Abdelrahman

Hassan

Omar

Mentor Review

Proposal Summary

Project identity

- ▶ Team: Abdelrahman, Hassan, Omar.
- ▶ Mentor: Kerven Durdymyradov.
- ▶ Track: AI, speech processing, SSL.

Core deliverable

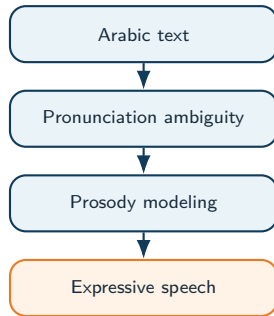
- ▶ Compact Audio-JEPA speech encoder.
- ▶ Low-resource Arabic TTS prototype.
- ▶ Baseline comparison and demo samples.

This is a proposal for a **research-backed prototype**, not only an application demo.

1. Background and Motivation

Why Arabic TTS is hard

- ▶ High-quality paired text–audio data is limited.
- ▶ Arabic has dialect variation and code-switching.
- ▶ Unvocalized text creates pronunciation ambiguity.
- ▶ Natural speech requires rhythm, pitch, energy, and duration modeling.



2. Problem Statement

Arabic TTS systems often need substantial clean paired text–audio data to generate natural speech. For many Arabic use cases, especially dialectal or expressive synthesis, this paired data is scarce.

Research Question

Can Audio-JEPA-style self-supervised speech representations improve the quality or data efficiency of low-resource Arabic TTS?

3. Objectives

Project objectives

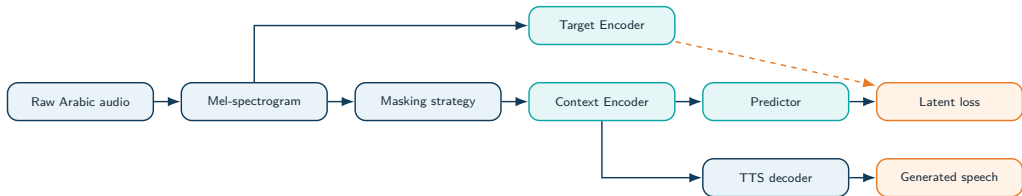
1. Build a clean Arabic audio preprocessing pipeline using log Mel-spectrograms.
2. Implement a compact Audio-JEPA-style self-supervised encoder.
3. Use limited paired Arabic text–audio data for TTS adaptation.
4. Compare against a small-data baseline TTS setup.
5. Deliver final prototype, report, slides, and audio demo samples.

4. Proposed Approach

The project separates **speech representation learning** from **speech generation**. First, the encoder learns from unlabeled Arabic audio. Then, the learned representations support low-resource TTS adaptation.



5. System Architecture



Training objective: latent prediction plus representation regularization.

$$\mathcal{L} = \lambda_1 \|z_p - z_t\|_2^2 + \lambda_2 \mathcal{L}_{var} + \lambda_3 \mathcal{L}_{cov}$$

6. Data Plan

Stage	Data Type	Purpose
Pre-training	Unlabeled Arabic audio	Learn phonetic, rhythmic, and prosodic representations
Fine-tuning	Paired Arabic text–audio	Train or condition the TTS decoder
Evaluation	Held-out Arabic sentences	Compare intelligibility, naturalness, and data efficiency

Safe MVP choice: Start with clean MSA data first; add dialectal or mixed Arabic audio only as a stretch goal.

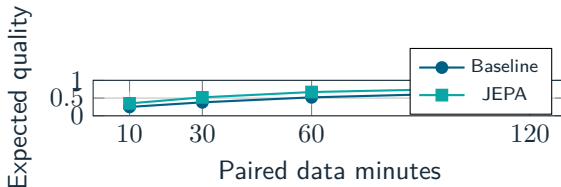
7. Evaluation Plan

Research evaluation

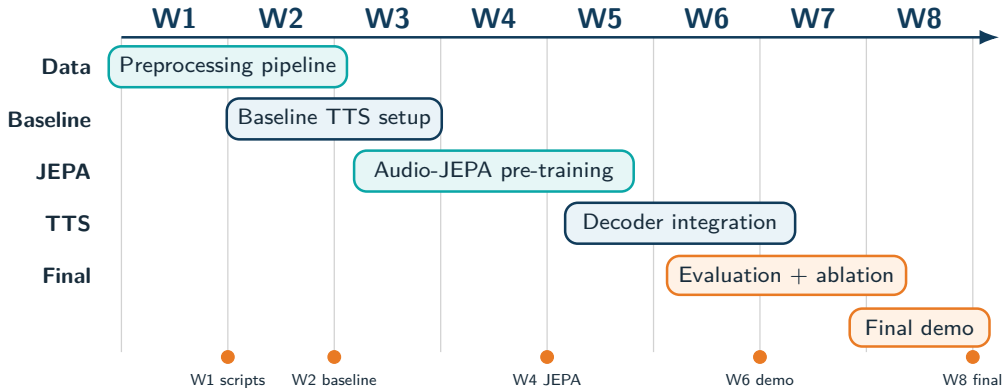
- ▶ Latent prediction loss.
- ▶ Speaker or dialect probing tasks.
- ▶ Embedding visualization or clustering.
- ▶ Baseline encoder comparison.

TTS evaluation

- ▶ Naturalness listening test.
- ▶ Intelligibility rating.
- ▶ Pronunciation and rhythm analysis.
- ▶ Data-efficiency comparison.



8. Work Plan and Timeline



9. Team Roles

Student	Primary responsibilities
Abdelrahman	Model architecture, Audio-JEPA design, training objective, research framing
Hassan	Data collection, preprocessing, dataset documentation, audio feature pipeline
Omar	Training loop, TTS decoder integration, benchmarking, demo preparation

Each member owns a distinct technical layer while keeping the final demo integrated.

10. Risks and Mitigation

Risk	Mitigation
Limited compute for full pre-training	Start with a compact model and reduced audio subset
Arabic preprocessing complexity	Build normalization and feature extraction scripts in week 1
Limited paired TTS data	Use a small-data baseline and evaluate data efficiency directly
Decoder integration difficulty	Keep a fallback demo using simpler conditioning or baseline synthesis

11. Expected Deliverables

Final outputs

- ▶ Arabic audio preprocessing pipeline.
- ▶ Compact Audio-JEPA-style self-supervised encoder.
- ▶ Low-resource Arabic TTS prototype or proof-of-concept integration.
- ▶ Baseline versus JEPA-conditioned comparison.
- ▶ Final report, presentation slides, and demo audio samples.

Success criterion: demonstrate whether SSL speech representations improve Arabic TTS quality or data efficiency.

12. Mentor Approval Questions

Questions for Mentor Approval

1. Is this scope acceptable for the KASP 2026 AI project timeline?
2. Should the MVP start with MSA-only, dialect subset, or mixed Arabic audio?
3. If time becomes limited, should the project prioritize research evidence, prototype quality, or both equally?
4. Are any compute resources or mentor-approved datasets available?
5. Is the proposed role split suitable for execution and weekly progress reviews?

Approve Audio-JEPA for Expressive, Low-Resource Arabic TTS

Why: research-rich, Arabic-relevant, compute-aware, and prototype-friendly.

MVP: compact Audio-JEPA-style encoder plus small-data Arabic TTS demo.

Next step: mentor approval, dataset selection, and week-1 preprocessing implementation.

Reading List

-  LeCun, Y. A Path Towards Autonomous Machine Intelligence. Open discussion essay, 2022.
-  Baevski, A. et al. data2vec: A General Framework for Self-supervised Learning in Speech, Vision and Language. ICML, 2022.
-  Hsu, W.-N. et al. HuBERT: Self-Supervised Speech Representation Learning by Masked Prediction of Hidden Units. IEEE/ACM TASLP, 2021.
-  Bardes, A., Ponce, J., LeCun, Y. VICReg: Variance-Invariance-Covariance Regularization for Self-Supervised Learning. ICLR, 2022.
-  Ren, Y. et al. FastSpeech 2: Fast and High-Quality End-to-End Text to Speech. ICLR, 2021.
-  Kong, J., Kim, J., Bae, J. HiFi-GAN: Efficient and High Fidelity Speech Synthesis. NeurIPS, 2020.